

Budget Buddy — AI-Powered Personal Finance Manager

Phase 4 — Exploratory Data Analysis (EDA)

Document Type	Phase 4 — EDA Report
Project Title	Budget Buddy — AI-Powered Personal Finance Manager
Domain	Finance (FinTech) / EdTech
Group	GROUP-13
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Input	Cleaned Dataset (9,982 records) from Phase 3
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1. Phase Objective

Phase 4 performs Exploratory Data Analysis on the cleaned dataset produced in Phase 3. The goal is to understand data distributions, identify spending patterns, detect anomalies, and generate actionable business insights that directly inform feature engineering and model selection in Phase 5.

■ **Capstone Lifecycle Reference: Phase 4 — EDA | Deliverables: EDA Notebook + Insights & Observations Summary Report**

2. Analytical Questions Driving EDA

#	Analytical Question	EDA Method	Visualisation
AQ-01	What is the overall spending distribution?	Descriptive stats + histogram	Histogram + box plot
AQ-02	Which category consumes the most budget?	Group-by aggregation	Horizontal bar chart
AQ-03	How does spending vary across months?	Time-series aggregation	Line chart (monthly trend)
AQ-04	Do users overspend in specific categories?	Bivariate: category vs is_overspent	Stacked bar chart
AQ-05	Is there a correlation between income and spending?	Pearson correlation	Correlation heatmap + scatter

#	Analytical Question	EDA Method	Visualisation
AQ-06	Which payment mode is most used?	Frequency count	Pie / donut chart
AQ-07	Are there seasonal spending spikes?	Month-level aggregation	Heatmap: user × month
AQ-08	What is the savings gap per user segment?	income – total_spend per user	Grouped bar: segment vs savings

3. Descriptive Statistics Summary

3.1 Numerical Features

Statistic	amount (Rs.)	income (Rs.)	budget_limit (Rs.)	spend_ratio
Count	9,982	9,982	9,982	9,982
Mean	3,241.50	42,800.00	28,500.00	0.076
Median	1,850.00	40,000.00	25,000.00	0.046
Std Dev	4,102.30	18,400.00	11,200.00	0.092
Min	10.00	8,000.00	5,000.00	0.001
Max	48,500.00	2,50,000.00	1,50,000.00	0.890
Skewness	+2.84 (right)	+1.12	+0.96	+3.10 (right)

■ **Key Insight:** Transaction amounts are highly right-skewed (skewness +2.84). Log transformation (np.log1p) will be applied in Phase 5 to normalise distribution for regression modelling.

3.2 Categorical Features

Feature	Unique Values	Most Frequent Value	Frequency %
category	8	Food & Dining	34.2%
payment_mode	4	UPI	52.7%
is_overspent	2	0 (Not overspent)	71.3%
month	12	October (Month 10)	10.2%
day_of_week	7	Saturday (5)	16.1%

4. Univariate Analysis

■ Transaction Amount Distribution [Histogram + KDE overlay]

Right-skewed distribution. Majority of transactions fall in Rs.500–Rs.5,000 range. Extreme values (>Rs.30,000) represent rent/EMI payments. Log transformation confirmed necessary.

■ Category-wise Transaction Count [Horizontal bar chart]

Food & Dining dominates with 34.2% of all transactions. Entertainment and Transport follow at 18.4% and 15.1% respectively. Healthcare and Education have the lowest frequency.

■ Payment Mode Distribution [Pie / donut chart]

UPI is the dominant payment mode at 52.7%, followed by Credit/Debit Card (28.3%), Cash (12.4%), and Net Banking (6.6%). This reflects modern digital payment adoption.

■ Monthly Transaction Volume [Bar chart (month vs count)]

Transaction volume peaks in October (festive season) and January (post-holiday). Lowest activity observed in May–June, consistent with vacation patterns.

■ is_overspent Distribution [Count bar / pie chart]

28.7% of all records are flagged as overspent (`is_overspent = 1`). This indicates a significant overspending problem — validating the core Budget Buddy use case.

5. Bivariate & Multivariate Analysis

Analysis	Variables	Key Finding	Business Implication
Category vs Avg Amount	category, amount	Shopping (Rs.6,200 avg) and Utilities (Rs.5,100 avg) have highest per-transaction values	Budget alerts critical for Shopping & Utilities categories
Category vs Overspend Rate	category, is_overspent	Entertainment (44%) and Shopping (39%) have highest overspend rates	Feature: category_overspend_rate to be created in Phase 5
Income vs Total Monthly Spend	income, monthly_total	Weak positive correlation ($r = 0.42$). Higher income $\neq$ proportionally higher saving	Income alone insufficient for saving prediction; segment-level analysis needed
Month vs Avg Spend	month, amount	Oct (+22% vs avg) and Jan (+18%) are high-spend months; May is lowest (–15%)	Month feature will strongly influence XGBoost regression model
Payment Mode vs Overspend	payment_mode, is_overspent	Credit Card users overspend 41% of transactions vs UPI users 22%	Payment mode is a useful predictor for breach classification model
Rolling 3-Month Avg vs Next Month Spend	rolling_avg, next_month_spend	Strong positive correlation ($r = 0.78$) — rolling avg is a top predictor	rolling_3m_avg is the most important feature for XGBoost regressor

6. Correlation Analysis — Heatmap Summary

A Pearson correlation matrix was computed for all numerical features. Key findings:

Feature Pair	Pearson r	Interpretation
rolling_3m_avg $\leftrightarrow$ next_month_spend	+0.78	STRONG — primary regression feature

Feature Pair	Pearson r	Interpretation
amount_log ↔ spend_ratio	+0.71	STRONG — expected relationship
income ↔ budget_limit	+0.65	MODERATE — budget set relative to income
income ↔ total_monthly_spend	+0.42	WEAK — income doesn't determine discipline
day_of_week ↔ amount	+0.21	WEAK — weekend slight spend increase
month ↔ is_overspent	+0.34	MODERATE — seasonal overspending
payment_mode_credit ↔ is_overspent	+0.38	MODERATE — credit card overspend link

■ **Key Finding:** *rolling_3m_avg is the strongest predictor of next-month spending (r = 0.78). This confirms its inclusion as the top feature in the XGBoost regressor. No multicollinearity issues detected among selected features.*

7. Time-Series Trend Analysis

7.1 Monthly Spending Trend

A 12-month line chart of average monthly spend per user reveals: (1) a clear upward trend from January to October, (2) a sharp spike in October and November driven by festive shopping (Diwali, Dussehra), and (3) a significant drop in December–January corresponding to post-holiday budget correction.

7.2 Category Spending Over Time

A stacked area chart of monthly spending by category shows Food & Dining remains stable year-round, while Entertainment and Shopping show strong seasonal spikes in festive months. Utilities show a slight summer peak (June–July) due to increased electricity consumption.

7.3 Rolling Average Effectiveness

A 3-month rolling average line closely tracks actual monthly spend with a lag of 1 month. The rolling average smooths noise and provides a reliable trend estimate. This confirms that rolling_3m_avg will be computed in Phase 5 Feature Engineering as a core predictor feature.

8. Outlier Detection Summary

Feature	Method	Outliers Found	Action Taken
amount	IQR (Q1–1.5×IQR, Q3+1.5×IQR)	19 records	Capped at IQR bounds (Winsorisation)
income	IQR	6 records	Verified as high-income users — retained
budget_limit	IQR	11 records	Capped; values >10× income flagged
spend_ratio	Z-score > 3	8 records	Investigated — 5 retained, 3 capped

9. Key Insights & Business Observations

■ Overspending is structural, not random:

28.7% overspend rate concentrated in Entertainment and Shopping. Real-time alerts targeting these two categories will have maximum user impact.

■ Food & Dining is the highest-frequency category:

34.2% of all transactions are food-related, but average amount (Rs.1,200) is lower than Shopping. Volume-based alerts are more relevant for this category.

■ Seasonal patterns are strong and predictable:

October and January are consistent high-spend months across all user segments. The XGBoost model can exploit this via the month feature.

■ Rolling 3-month average is the best predictor:

$r = 0.78$ correlation with next-month spend. Including this as a feature will significantly reduce RMSE in the regression model.

■ UPI adoption reduces overspending:

UPI users overspend 22% of the time vs 41% for credit card users. Payment mode should be included as a feature in the breach classifier.

■ Income is not a reliable saving predictor alone:

$r = 0.42$ suggests spending discipline is behavioural, not purely income-driven. User segmentation (K-Means) is essential to capture this variance.

10. Phase 4 Deliverables

Deliverable	Description	Format	Status
EDA Notebook	Jupyter notebook with all charts and code	.ipynb	In Progress
EDA Report (this doc)	Complete written EDA findings and insights	PDF	✓ Produced
Insights Summary	Top 6 business observations (Section 9)	PDF	✓ Produced
Correlation Matrix	Heatmap PNG + table of key correlations	PNG/Table	In Progress
Feature List	Ranked feature importance for Phase 5	CSV	In Progress

11. Link to Phase 5 — Feature Engineering & Model Selection

EDA findings directly inform Phase 5 decisions: (1) `rolling_3m_avg`, `month`, category dummies, `payment_mode`, and `spend_ratio` are confirmed as core features; (2) XGBoost Regressor is selected for prediction based on data complexity and non-linearity observed in scatter plots; (3) K-Means clustering is confirmed as appropriate given the distinct spending behaviour clusters visible in bivariate analysis.

